

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	July 8 2026
Team ID	SWTID-2026-2791
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Crop Recommendation	Recommend suitable crops, Display recommended crop details
FR-4	Soil Analysis	Enter soil parameters, Analyze soil health
FR-5	Weather Monitoring	Display weather forecast, Generate weather alerts
FR-6	Yield Prediction	Predict crop yield, Show production estimation

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Easy-to-use interface for farmers
NFR-2	Security	Secure user authentication and data protection
NFR-3	Reliability	Generate recommendations within a few seconds
NFR-4	Performance	Provide accurate and consistent predictions
NFR-5	Availability	System should be available 24/7
NFR-6	Scalability	Support increasing users and agricultural data